

1. The speed of a wave in deep water depends on its wavelength L and the acceleration of free fall g .

What is a possible equation for the speed v of the wave?

A $v = \frac{2\pi g}{L}$

B $v = \frac{gL}{4\pi^2}$

C $v = 2\pi\sqrt{\frac{g}{L}}$

D $v = \sqrt{\frac{gL}{2\pi}}$

Answer: D

Units for the RHS equals to units of v (m s^{-1})

A: $\frac{\text{m s}^{-2}}{\text{m}} = \text{s}^{-2}$

B: $\text{m s}^{-2} \square \text{m} = \text{m}^2 \text{s}^{-2}$

C: $\left(\frac{\text{m s}^{-2}}{\text{m}}\right)^{1/2} = \text{s}^{-1}$

D: $\left(\text{m s}^{-2} \square \text{m}\right)^{1/2} = \text{m s}^{-1}$

2. The power P dissipated in a resistor of resistance R is calculated using the expression